8.1.3 Tweaking performance

Performance testing of hard disks is divided into read and write tests. To test the performance of your volumes, you will use a simple tool named `dd`, which can perform block-level reads and writes between a source `if=/path/to/source` and a destination `of=/path/to/destination`, shown next. For comparison, you'll run a performance test for a temporary block-level storage volume in the following section:

```
$ sudo dd if=/dev/zero of=/data/tempfile bs=1M count=1024 \      ①
➥ conv=fdatasync,notrunc
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB) copied, 15.8331 s, 67.8 MB/s           ②

$ echo 3 | sudo tee /proc/sys/vm/drop_caches                     ③
3
$ sudo dd if=/data/tempfile of=/dev/null bs=1M                   ④
➥ count=1024
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB) copied, 15.7875 s, 68.0 MB/s           ⑤
```

① Writes 1 MB 1,024 times

② 67.8 MB/s write performance

③ Flushes caches

④ Reads 1 MB 1,024 times

⑤ 68.0 MB/s read performance

Keep in mind that performance can be different depending on your actual workload. This example assumes that the file size is 1 MB. If you're hosting websites, you'll most likely deal with lots of small files instead.